



FINAL MARK

**GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION 2
TASK 2 2019 – Polynomials and Conic Sections
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	E2	E3	E4	E5	E6	E7	E8	E9
MC	/5			✓					✓
Q6	/18			✓					✓
Q7	/16		✓	✓					✓
Q8	/9		✓	✓					✓
Q9	/8		✓	✓					✓
TOTAL									
	/56	/0	/33	/56	/0	/0	/0	/0	/56

- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.



GIRRAWEEEN HIGH SCHOOL

MATHEMATICS EXTENSION 2

2019 HSC Task 2

MULTIPLE CHOICE ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☐
3.	A	☐	B	☐	C	☐	D	☐
4.	A	☐	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☐



GIRRAWEE HIGH SCHOOL

Task 2

2019

MATHEMATICS

EXTENSION 2

Time Allowed: 90 minutes

DIRECTIONS TO CANDIDATES

- There are 9 questions in this paper. All questions are compulsory.
- Use blue or black pen.
- Write all your answers in the Answer Booklets provided.
- For Questions 1 – 5, fill in the circle corresponding to the correct answer in your answer booklet.
- For Questions 6 – 9, start each question on a new page.
- Write on both sides of the paper.
- Show all necessary working.
- Board-approved calculators may be used.
- Mathematics reference sheets are provided.
- Marks may be deducted for careless or badly arranged work.

Questions 1 - 5 (5 marks)

Colour in the circle corresponding to the correct answer on the answer sheet provided.

1. What are the values of real numbers p and q such that $1 - i$ is a root of the equation $z^3 + pz + q = 0$?

(A) $p = -2$ and $q = -4$
(B) $p = -2$ and $q = 4$
(C) $p = 2$ and $q = 4$
(D) $p = 2$ and $q = -4$

2. What is the double root of the equation $x^3 - 5x^2 + 8x - 4 = 0$?

(A) $x = -2$
(B) $x = -1$
(C) $x = 1$
(D) $x = 2$

3. For the ellipse with the equation $\frac{(x-2)^2}{5} + \frac{(y+3)^2}{3} = 1$, what is the eccentricity?

(A) $\sqrt{\frac{2}{5}}$

(B) $\sqrt{\frac{2}{3}}$

(C) $\sqrt{\frac{5}{3}}$

(D) $\sqrt{\frac{5}{2}}$

4. The equations of the asymptotes of the hyperbola $9x^2 - 4y^2 = 36$ are

(A) $x = \pm \frac{3}{2}y$

(B) $y = \pm \frac{3}{2}x$

(C) $x = \pm \frac{9}{4}y$

(D) $y = \pm \frac{9}{4}x$

5. The equation $x^3 - y^3 + 3xy + 1 = 0$ defines y implicitly as a function of x .

Which of the following is the expression for $\frac{dy}{dx}$?

(A) $\frac{y^2 - x}{x^2 + y}$

(B) $\frac{y^2 + x}{x^2 - y}$

(C) $\frac{x^2 + y}{y^2 - x}$

(D) $\frac{x^2 - y}{y^2 + x}$

Question 6 (18 marks)

a. If α , β and γ are the roots of the cubic equation $x^3 + mx + n = 0$,

(i) find, in terms of m and n , the values of:

I. $\alpha + \beta + \gamma$ [1]

II. $\alpha\beta + \beta\gamma + \alpha\gamma$ [1]

III. $\alpha\beta\gamma$ [1]

IV. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]

V. $\alpha^3 + \beta^3 + \gamma^3$ [2]

(ii) Determine the cubic equation whose roots are α^2 , β^2 and γ^2 [3]

b. The polynomial $P(x) = x^4 + 7x^3 + 9x^2 - 27x + C$ has a triple zero.

(i) Determine the value of the triple zero. [2]

(ii) Hence, find the value of C . [1]

(iii) Factorise $P(x)$. [1]

c. If α is a non-real double root of $P(x) = x^4 - 4x^3 + 14x^2 - 20x + 25$, factorise $P(x)$ completely into linear factors. [4]

Question 7 (16 marks)

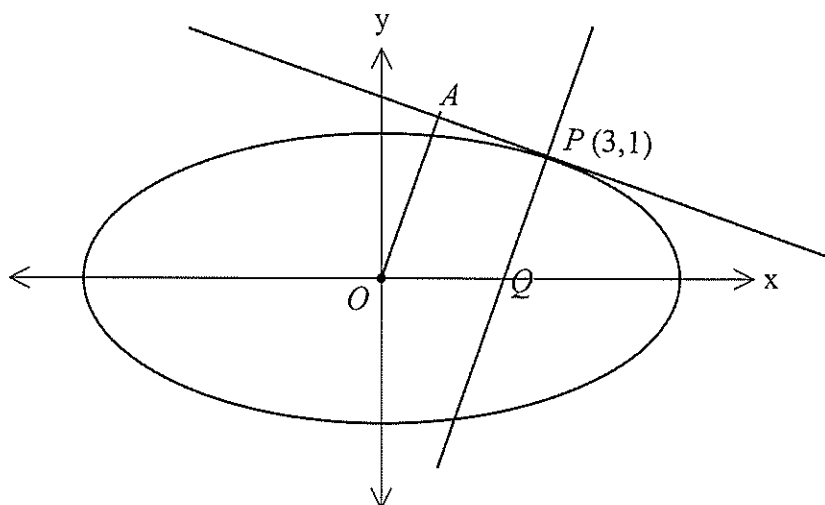
a. The equation of an ellipse is given by $\frac{x^2}{4} + \frac{y^2}{3} = 1$

- (i) Find the eccentricity. [1]
- (ii) Find the coordinates of the foci, S and S' . [1]
- (iii) Find the equations of the directrices. [1]
- (iv) Sketch the ellipse, clearly showing the important features. [2]
- (v) Let P be any point on the ellipse. Show that the sum of the distances SP and $S'P$ is independent of P . [2]
- (vi) Find the equation of the chord of contact from $(3, 2)$ [2]

[Equation of chord of contact: $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$]

b. An ellipse has the equation $x^2 + 16y^2 = 25$

- (i) Find the equations of the **tangent** and **normal** to the ellipse at the point $P(3, 1)$. [4]
- (ii) The normal in (i) meets the major axis at Q .
A line from the origin, O , meets the tangent in (i) at right angles at A .
Show that the value of $PQ \times OA$ is equal to the square of the semi-minor axis. [3]



Question 8 (9 marks)

Consider the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$.

- a. Sketch the graph of the hyperbola, clearly showing any intercepts with the coordinate axes, the coordinates of the foci and the equations of the directrices and asymptotes. [3]
- b. $T(x_1, y_1)$ is a point on the hyperbola.
Show that the equation of the tangent to the hyperbola at T is $\frac{xx_1}{4} - \frac{yy_1}{12} = 1$. [3]
- c. $P(4, 6)$ and $Q(14, 24)$ are two points on the hyperbola.
 M is the midpoint of PQ and $O(0, 0)$ is the origin.
The tangents at P and Q intersect at R .
Show that the points R , O and M are collinear. [3]

Question 9 (8 marks)

Consider the hyperbola with the equation $xy = 25$.

- a. $P\left(5p, \frac{5}{p}\right)$ and $Q\left(5q, \frac{5}{q}\right)$ are two distinct points on the hyperbola and p and q are both greater than zero.
Find the equation of the chord PQ . [2]
- b. Prove that the equation of the tangent at P is $x + p^2y = 10p$. [2]
- c. The tangents at P and Q intersect at T .
Find the coordinates of T . [2]
- d. The chord PQ produced passes through the point $N(0, 10)$.
Find and describe the locus of T . [2]

End of Examination

2019 GHS EXT 2 TASK 2 SOLUTIONS.

MC (5 marks)

1. real coefficients $\therefore 1+i$ is also a root
roots: $1 \pm i, \alpha$

$$\text{sum of roots} \Rightarrow 2 + \alpha = 0$$

$$\alpha = -2$$

$$\text{Product of roots} \Rightarrow (1+i)(1-i)x - 2 = -q$$

$$q = 4$$

$$\Delta = (1+i)(1-i) + -2(1+i) + -2(1-i) - p$$

$$p = -2$$

[B]

2. $f'(x) = 3x^2 - 10x + 8$

$$(3x - 4)(x - 2) = 0$$

$$x = \frac{4}{3}, 2$$

[D]

3. $b^2 = a^2(1 - e^2)$

$$3 = 5(1 - e^2)$$

$$e^2 = \frac{2}{5}$$

$$e = \sqrt{\frac{2}{5}}$$

[A]

4. $\frac{x^2}{4} - \frac{y^2}{9} = 1 \quad a=2, b=3$

Asymptotes: $y = \pm \frac{b}{a}x$

$$= \pm \frac{3}{2}x$$

[B]

5. $3x^2 - 3y^2 \frac{dy}{dx} + 3x \frac{dy}{dx} + 3y = 0$

$$\frac{dy}{dx} (3x - 3y^2) = -3x^2 - 3y$$

$$\frac{dy}{dx} = \frac{-3(x^2 + y)}{-3(y^2 - x)} = \frac{x^2 + y}{y^2 - x}$$

[C]

Question 6 (18 marks)

a) $x^3 + mx + n = 0$

Roots : α, β, γ

i) I. $\alpha + \beta + \gamma = 0$

II. $\alpha\beta + \beta\gamma + \alpha\gamma = m$

III. $\alpha\beta\gamma = -n$

IV. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$$= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$= -\frac{m}{n}$$

V. $\alpha^3 + \beta^3 + \gamma^3$

If α, β, γ are roots, then

$$\alpha^3 + m\alpha + n = 0 \quad \text{--- (1)}$$

$$\beta^3 + m\beta + n = 0 \quad \text{--- (2)}$$

$$\gamma^3 + m\gamma + n = 0 \quad \text{--- (3)}$$

(1) + (2) + (3)

$$\alpha^3 + \beta^3 + \gamma^3 + m(\alpha + \beta + \gamma) + 3n = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = -3n$$

ii) $y = x^2 \Rightarrow x = \sqrt{y}$

substituting,

$$(\sqrt{y})^3 + m\sqrt{y} + n = 0$$

$$y\sqrt{y} + m\sqrt{y} + n = 0$$

$$\sqrt{y}(y + m) = -n$$

$$y(y^2 + 2my + m^2) = n^2$$

squaring both sides

$$y^3 + 2my^2 + m^2y - n^2 = 0$$

or $x^3 + 2mx^2 + m^2x - n^2 = 0$

$$6b) P(x) = x^4 + 7x^3 + 9x^2 - 27x + C \quad \leftarrow \text{triple zero}$$

$$i) P'(x) = 4x^3 + 21x^2 + 18x - 27 \quad \leftarrow \text{double of same.}$$

$$P''(x) = 12x^2 + 42x + 18 \quad \leftarrow \text{single}$$

$$P''(x) = 0$$

$$2x^2 + 7x + 3 = 0$$

$$(2x+1)(x+3) = 0$$

$$x = -\frac{1}{2}, -3$$

$$P'(-3) = 4(-3)^3 + 21(-3)^2 + 18(-3) - 27 = 0$$

$\therefore$ triple zero at $x = -3$

$$ii) P(-3) = 0$$

$$(-3)^4 + 7(-3)^3 + 9(-3)^2 - 27(-3) + C = 0$$

$$C = -54$$

$$iii) P(x) = (x+3)^3 \cdot Q(x)$$

$$P(x) = (x+3)^3 (x-2) \quad [\text{By observation}]$$

$$(C = -54)$$

$$c) P(x) = x^4 - 4x^3 + 14x^2 - 20x + 25,$$

If α is a double root, then $\bar{\alpha}$ is also double root.
 $\therefore$ roots : $\alpha, \alpha, \bar{\alpha}, \bar{\alpha}$

$$\text{Sum of roots : } \alpha + \alpha + \bar{\alpha} + \bar{\alpha} = 4$$

$$\text{let } \alpha = a + ib \Rightarrow 4a = 4$$

$$a = 1$$

$$\text{Product of roots : } \alpha \alpha \bar{\alpha} \bar{\alpha} = 25 \quad \text{and} \quad a^2 + b^2 = 5$$

$$(\alpha \bar{\alpha})^2 = 25$$

$$1 + b^2 = 5$$

$$\alpha \bar{\alpha} = 5$$

$$b = \pm 2$$

$$\therefore P(x) = (x-1+2i)^2 (x-1-2i)^2$$

Question 7 (16 marks)

a) $\frac{x^2}{4} + \frac{y^2}{3} = 1$

$$a^2 = 4, b^2 = 3$$

i) $b^2 = a^2(1 - e^2)$

(ii) Foci $(\pm ae, 0)$

$$3 = 4(1 - e^2)$$

$$e^2 = \frac{1}{4}$$

$$S(1, 0), S'(-1, 0)$$

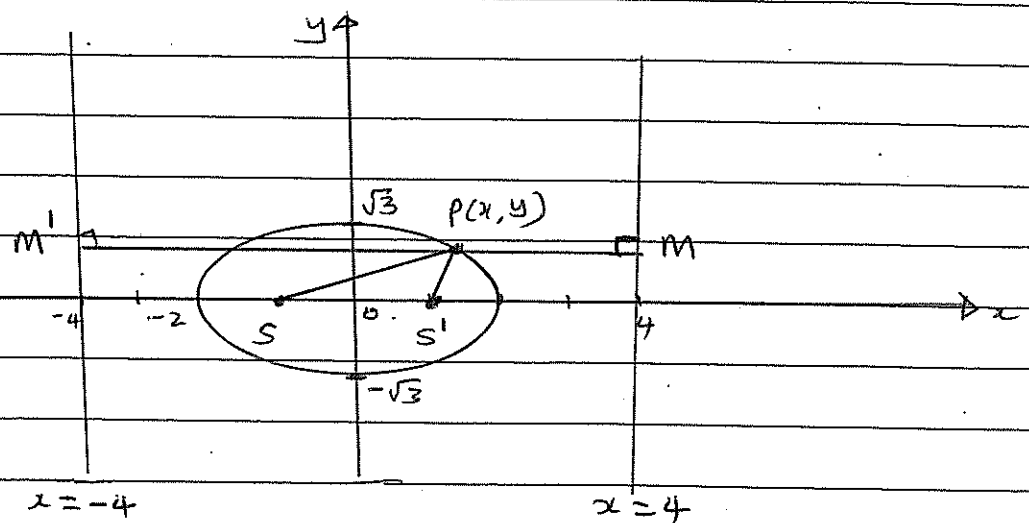
$$e = \frac{1}{2}$$

iii) Directrices: $x = \pm \frac{a}{e}$

$$= \pm \frac{2}{\frac{1}{2}}$$

$$x = \pm 4$$

iv)



v) $PS = e PM$ — (1)
 $PS' = e PM'$ — (2) } definition of ellipse

(1) + (2)

$$PS + PS' = e(PM + PM')$$

$$= e(8)$$

$$= \frac{1}{2}(8) = 4$$

$\therefore PS + PS' = 4$ which is independent of P.

7a) vi) $E_{\text{CHORD OF CONTACT}}$

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1 \quad (x_0, y_0) = (3, 2)$$

$$\frac{3x}{4} + \frac{2y}{3} = 1$$

$$9x + 8y - 12 = 0$$

7b) $x^2 + 16y^2 = 25$

i) $2x + 32y \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{-x}{16y}$$

At $P(3, 1)$, $m_{\text{tangent}} = \frac{-3}{16}$; $m_{\text{normal}} = \frac{16}{3}$

$$E_{\text{tangent}} : y - 1 = \frac{-3}{16} (x - 3)$$

$$16y - 16 = -3x + 9$$

$$3x + 16y - 25 = 0$$

$$E_{\text{normal}} : y - 1 = \frac{16}{3} (x - 3)$$

$$16x - 3y - 45 = 0$$

$$7b \text{ (ii)} \quad \frac{x^2}{25} + \frac{16y^2}{25} = 1$$

$$b^2 = \frac{25}{16} \quad \leftarrow \begin{array}{l} \text{Square of} \\ \text{semi-minor} \\ \text{axis} \end{array}$$

$$OA = \frac{|3(0) + 4(0) - 25|}{\sqrt{3^2 + 16^2}}$$

$$P(3, 1)$$

$$= \frac{25}{\sqrt{265}}$$

PQ meets the major axis when $y = 0$

$$\therefore 16x = 45$$

$$x = \frac{45}{16}$$

$$Q\left(\frac{45}{16}, 0\right)$$

$$PQ = \sqrt{\left(3 - \frac{45}{16}\right)^2 + (1 - 0)^2}$$

$$= \frac{\sqrt{265}}{16}$$

$$\therefore PQ \times OA = \frac{\sqrt{265}}{16} \times \frac{25}{\sqrt{265}}$$

$$= \frac{25}{16}$$

$$= b^2 \text{ (square of semi-minor axis)}$$

Q8 (9 marks)

$$\frac{x^2}{4} - \frac{y^2}{12} = 1$$

$$a^2 = 4; b^2 = 12$$

$$a) 12 = 4(e^2 - 1)$$

$$e^2 = 4$$

$$e = 2$$

Foci $(\pm ae, 0)$

$$(\pm 4, 0)$$

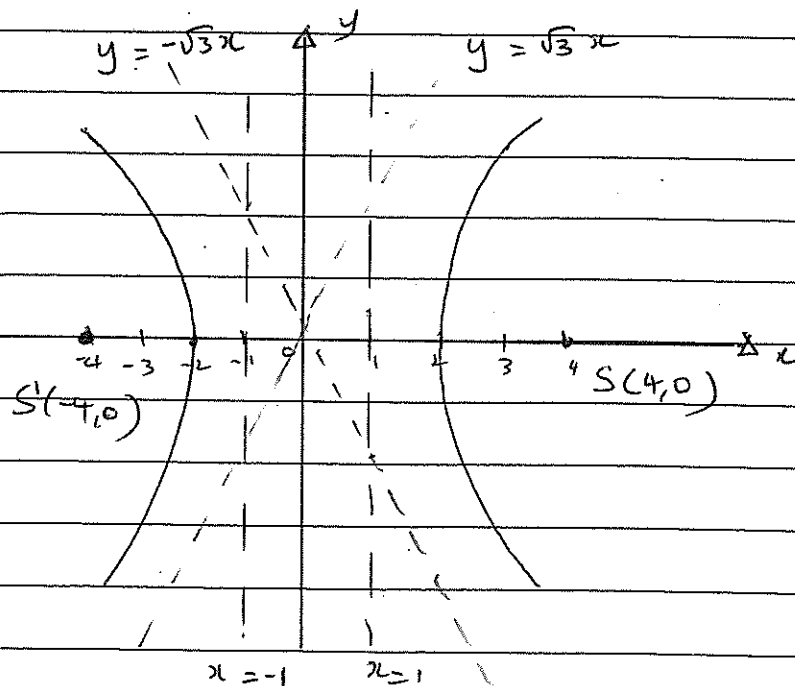
$$\text{Directrices: } x = \pm \frac{a}{e}$$

$$x = \pm 1$$

$$\text{Asymptotes: } y = \pm \frac{b}{a} x$$

$$= \pm \frac{\sqrt{12}}{2} x$$

$$y = \pm \sqrt{3} x$$



$$8b) \quad \frac{x^2}{4} - \frac{y^2}{12} = 1$$

$$\frac{x}{2} - \frac{y}{6} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{3x}{y}$$

$$\text{At } T(x_1, y_1), \quad m_{\text{tangent}} = \frac{3x_1}{y_1}$$

$$E_{\text{tangent}} : y - y_1 = \frac{3x_1}{y_1} (x - x_1)$$

$$yy_1 - y_1^2 = 3xx_1 - 3x_1^2$$

$$3xx_1 - yy_1 = 3x_1^2 - y_1^2 \quad (\div 12)$$

$$\frac{xx_1}{4} - \frac{yy_1}{12} = \frac{x_1^2}{4} - \frac{y_1^2}{12}$$

$$(T \text{ lies on hyperbola} \\ \therefore \frac{x_1^2}{4} - \frac{y_1^2}{12} = 1)$$

$$\therefore \frac{xx_1}{4} - \frac{yy_1}{12} = 1$$

$$c) \quad P(4, 6), \quad Q(14, 24)$$

$$M = (9, 15)$$

$$\text{Tangent at } P : \frac{4x}{4} - \frac{6y}{12} = 1 \Rightarrow x - \frac{y}{2} = 1 \quad \text{--- (1)}$$

$$\text{Tangent at } Q : \frac{14x}{4} - \frac{24y}{12} = 1 \Rightarrow \frac{7x}{2} - 2y = 1 \quad \text{--- (2)}$$

Tangents intersect at R

Solving (1) and (2) simultaneously $\Rightarrow 4 \times (1) - (2)$

$$\begin{array}{r} 4x - 2y = 4 \\ - \quad \frac{7x}{2} - 2y = 1 \\ \hline \end{array}$$

$$\frac{x}{2} = 3$$

$$x = 6$$

$$y = 10$$

$$\therefore R = (6, 10)$$

$$m_{OM} = \frac{15}{9} ; m_{OR} = \frac{5}{3} \\ = \frac{5}{3}$$

$\therefore R, O, M$ are collinear.

(8)

Q9 (8 marks)

a) $P(5p, \frac{5}{p})$; $Q(5q, \frac{5}{q})$

$$m_{PQ} = \frac{\frac{5}{q} - \frac{5}{p}}{5q - 5p} = \frac{5\left(\frac{p-q}{qp}\right)}{5(q-p)} \\ = -\frac{1}{pq}$$

$$E_{PQ} : y - \frac{5}{p} = -\frac{1}{pq}(x - 5p)$$

$$pqy - 5q = -x + 5p$$

$$x + pqy - 5(p+q) = 0$$

b) $xy = 25$

$$y = \frac{25}{x}$$

$$\frac{dy}{dx} = -\frac{25}{x^2}$$

A + $P(5p, \frac{5}{p})$, $m_{\text{tangent}} = -\frac{1}{p^2}$

$$E_{\text{tangent}} : y - \frac{5}{p} = -\frac{1}{p^2}(x - 5p)$$

$$p^2y - 5p = -x + 5p$$

$$x^2 + p^2y = 10p$$

$$c) E_{\text{tangent at } P} : x + p^2 y = 10p \\ \Rightarrow x = 10p - p^2 y \quad \text{--- (1)}$$

$$E_{\text{tangent at } Q} : x + q^2 y = 10q \\ \Rightarrow x = 10q - q^2 y \quad \text{--- (2)}$$

Solving (1) & (2) simultaneously

$$10p - p^2 y = 10q - q^2 y$$

$$q^2 y - p^2 y = 10q - 10p$$

$$y(q^2 - p^2) = 10(q - p)$$

$$y = \frac{10}{p+q}$$

$$x = 10p - p^2 \left(\frac{10}{p+q} \right)$$

$$= \frac{10p(p+q) - 10p^2}{p+q}$$

$$= \frac{10p^2 + 10pq - 10p^2}{p+q}$$

$$x = \frac{10pq}{p+q}$$

$$\therefore T = \left(\frac{10pq}{p+q}, \frac{10}{p+q} \right)$$

d) PQ passes through N(0, 10)

$\therefore$ N must satisfy the equation of chord PQ

$$x + pqy - 5(p+q) = 0$$

$$\therefore 10pq = 5(p+q)$$

$$pq = \frac{p+q}{2} \quad \text{--- (1)}$$

For T,

$$x = \frac{10pq}{p+q}$$

$$= \frac{10 \left(\frac{p+q}{2} \right)}{p+q} \quad (\text{from (1)})$$

$$= 5$$

$\therefore$ Locus is the vertical line

$$x = 5$$